

D115

Advanced Pathophysiology for the Advanced Practice Nurse

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 68-year-old retired machinist who enjoys fishing has a history of a large anterior heart attack. He now wakes at night short of breath, sleeps on three pillows, and has crackles at both lung bases. He takes a daily multivitamin. Which process best explains his symptoms?

- A. Left-sided heart failure backing blood into the pulmonary circulation.
- B. A vitamin-related fluid shift.
- C. Pneumonia, because crackles always mean infection.
- D. Right-sided heart failure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A patient with long-standing COPD develops jugular venous distention, an enlarged liver, and ankle edema, with clear lungs on auscultation. Which process fits?

- A. Hypovolemia.
- B. Left-sided heart failure.
- C. Liver failure alone.
- D. Right-sided heart failure (cor pulmonale).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 55-year-old who was sedated after surgery at a large teaching hospital has an ABG: pH 7.30, PaCO₂ 55 mmHg, HCO₃ 25 mEq/L. She has a history of seasonal allergies. What is the interpretation?

- A. Uncompensated respiratory acidosis.
- B. Fully compensated respiratory acidosis.
- C. Metabolic acidosis.
- D. Respiratory alkalosis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An ABG shows pH 7.31, PaCO₂ 30 mmHg, HCO₃ 15 mEq/L. What is the interpretation?

- A. Fully compensated metabolic acidosis.
- B. Respiratory acidosis.
- C. Metabolic acidosis with partial respiratory compensation.
- D. Respiratory alkalosis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A patient with small cell lung cancer, a former teacher who enjoys gardening, has a serum sodium of 124 mEq/L, low urine output, and highly concentrated urine. She is not taking diuretics. Which condition best fits?

- A. SIADH.
- B. Dehydration.
- C. Diabetes insipidus.
- D. Diuretic-induced hyponatremia.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

After a head injury, a patient passes several liters of very dilute urine a day and has a serum sodium of 151 mEq/L. Which condition fits?

- A. SIADH.
- B. Diabetes mellitus.
- C. Acute kidney injury.
- D. Diabetes insipidus.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A 70-year-old retired librarian who follows a strict vegan diet reports fatigue and numbness in her feet. Hemoglobin is low and MCV is 112 fL. She drinks herbal tea daily. What is the most likely cause?

- A. Tea blocking iron absorption.
- B. Vitamin B12 deficiency.
- C. Iron deficiency.
- D. Folate deficiency, because it is identical to B12 deficiency.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A patient three days after a large heart attack, who is a high school principal, has BP 82/50, cool clammy skin, crackles, and a high pulmonary capillary wedge pressure. He was admitted on a holiday weekend. What type of shock is this?

- A. Hypovolemic shock.
- B. Neurogenic shock.
- C. Septic shock.
- D. Cardiogenic shock.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A 72-year-old with type 2 diabetes who lives alone and volunteers at a library is brought in confused after days of poor fluid intake. Glucose 880 mg/dL, serum osmolality 350 mOsm/kg, pH 7.36, small ketones. Which condition fits?

- A. Diabetic ketoacidosis.
- B. Hyperosmolar hyperglycemic state (HHS).
- C. Hypoglycemia.
- D. Stroke, since confusion always means stroke.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A 19-year-old with type 1 diabetes has glucose 420 mg/dL, pH 7.18, HCO₃ 10 mEq/L, and large serum ketones. Which condition fits?

- A. Lactic acidosis from exercise.
- B. Diabetic ketoacidosis (DKA).
- C. HHS.
- D. Respiratory acidosis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A 28-year-old barista who runs marathons has episodic wheezing. Spirometry shows reduced FEV1 that improves by 18% after a bronchodilator. She has a cat. What pattern best fits?

- A.** Cat allergy only, so no airway disease.
- B.** Restrictive lung disease.
- C.** Asthma, with reversible airflow obstruction.
- D.** COPD.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A patient who works as a pilot has fatigue, low blood pressure, darkening skin, serum sodium 128 mEq/L, and potassium 5.9 mEq/L. He recently ran a half marathon. Which condition fits?

- A.** Cushing syndrome.
- B.** Exercise-related dehydration.
- C.** Primary adrenal insufficiency (Addison disease).
- D.** SIADH.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A 68-year-old retired machinist ~~who enjoys fishing~~ has a history of a large anterior heart attack. He now wakes at night short of breath, sleeps on three pillows, and has crackles at both lung bases. ~~He takes a daily multivitamin.~~ Which process best explains his symptoms?

Target word: side of failure **Decides:** Pulmonary congestion signs.

A	Correct	Orthopnea, nocturnal dyspnea, and crackles are pulmonary congestion.
B	Camouflage	The multivitamin does not decide the item.
C	Absolute language	Crackles have many causes; the pattern fits congestion.
D	Inverted operation	Right-sided failure causes systemic congestion such as JVD and leg edema.

2. Answer D Concept 1 • transfer

A patient with long-standing COPD develops jugular venous distention, an enlarged liver, and ankle edema, with clear lungs on auscultation. Which process fits?

Target word: side of failure **Decides:** Systemic venous congestion.

A	Confusion set	JVD and edema show fluid overload, not loss.
B	Inverted operation	Would cause pulmonary congestion; lungs are clear.
C	Adjacent category	Hepatomegaly here reflects venous congestion.
D	Correct	Chronic lung disease strains the right ventricle, causing systemic congestion.

3. Answer A Concept 2

A 55-year-old who was sedated after surgery ~~at a large teaching hospital~~ has an ABG: pH 7.30, PaCO₂ 55 mmHg, HCO₃ 25 mEq/L. ~~She has a history of seasonal allergies.~~ What is the interpretation?

Target word: ABG interpretation **Decides:** Low pH with high PaCO₂; HCO₃ normal.

A	Correct	pH low, PaCO ₂ high (opposite direction), HCO ₃ normal.
B	Dropped element	pH is still abnormal and HCO ₃ has not risen.
C	Confusion set	HCO ₃ is normal, so the cause is not metabolic.
D	Inverted operation	A high PaCO ₂ with low pH is acidosis.

4. Answer C Concept 2 • transfer

An ABG shows pH 7.31, PaCO₂ 30 mmHg, HCO₃ 15 mEq/L. What is the interpretation?

Target word: ABG interpretation **Decides:** Low pH, low HCO₃; PaCO₂ low as compensation.

A	Dropped element	pH has not returned to normal.
B	Inverted operation	PaCO ₂ is low, not high.
C	Correct	HCO ₃ is low with pH; PaCO ₂ falls to offset; pH still abnormal.
D	Confusion set	A low PaCO ₂ here is compensation; pH shows acidosis.

5. Answer A Concept 3

A patient with small cell lung cancer, ~~a former teacher who enjoys gardening~~, has a serum sodium of 124 mEq/L, low urine output, and highly concentrated urine. She is not taking diuretics. Which condition best fits?

Target word: condition **Decides:** Low sodium with concentrated urine.

A	Correct	Excess ADH retains water, diluting sodium and concentrating urine.
B	Confusion set	Dehydration usually raises sodium.
C	Inverted operation	DI causes dilute urine and high sodium.
D	Camouflage	The stem rules out diuretics.

6. Answer D Concept 3 · transfer

After a head injury, a patient passes several liters of very dilute urine a day and has a serum sodium of 151 mEq/L. Which condition fits?

Target word: condition **Decides:** High sodium with dilute urine.

A	Inverted operation	SIADH causes low sodium and concentrated urine.
B	Confusion set	Osmotic diuresis from glucose; the stem gives no hyperglycemia.
C	Adjacent category	Usually lowers urine output.
D	Correct	Lack of ADH effect causes large dilute urine and high sodium.

7. Answer B Concept 4

A 70-year-old retired librarian who follows a strict vegan diet reports fatigue and numbness in her feet. Hemoglobin is low and MCV is 112 fL. ~~She drinks herbal tea daily.~~ What is the most likely cause?

Target word: cause of anemia **Decides:** Macrocytic with neurologic signs and vegan diet.

A	Camouflage	Tea affects iron, which would not raise MCV.
B	Correct	Macrocytic anemia with neurologic signs fits B12 deficiency.
C	Inverted operation	Iron deficiency causes microcytic anemia.
D	Confusion set	Folate deficiency is macrocytic but lacks neurologic signs.

8. Answer D Concept 5

A patient three days after a large heart attack, ~~who is a high school principal~~, has BP 82/50, cool clammy skin, crackles, and a high pulmonary capillary wedge pressure. ~~He was admitted on a holiday weekend.~~ What type of shock is this?

Target word: shock type **Decides:** Pump failure with high filling pressures.

A	Inverted operation	Hypovolemia lowers filling pressures.
B	Adjacent category	Follows spinal cord injury with bradycardia.
C	Confusion set	Early septic shock often has warm skin and low SVR.
D	Correct	Low output with high filling pressure after MI.

9. Answer B Concept 6

A 72-year-old with type 2 diabetes ~~who lives alone and volunteers at a library~~ is brought in confused after days of poor fluid intake. Glucose 880 mg/dL, serum osmolality 350 mOsm/kg, pH 7.36, small ketones. Which condition fits?

Target word: crisis type **Decides:** Very high glucose and osmolality without acidosis.

A	Confusion set	DKA has significant ketosis and pH below 7.30.
B	Correct	Extreme hyperglycemia and osmolality, no significant acidosis.
C	Inverted operation	Glucose is very high.
D	Absolute language	Confusion here is explained by hyperosmolality.

10. Answer B Concept 6 · transfer

A 19-year-old with type 1 diabetes has glucose 420 mg/dL, pH 7.18, HCO₃ 10 mEq/L, and large serum ketones. Which condition fits?

Target word: crisis type **Decides:** Ketosis with metabolic acidosis.

A	Inverted operation	Large ketones point to DKA.
B	Correct	Insulin lack drives ketone production and acidosis.
C	Confusion set	HHS lacks significant acidosis and ketosis.
D	Adjacent category	Low HCO ₃ shows a metabolic cause.

11. Answer C Concept 7

A 28-year-old barista ~~who runs marathons~~ has episodic wheezing. Spirometry shows reduced FEV1 that improves by 18% after a bronchodilator. ~~She has a cat.~~ What pattern best fits?

Target word: asthma vs. COPD **Decides:** Reversible obstruction.

A	Camouflage	Triggers do not rule out asthma.
B	Adjacent category	Restriction reduces volumes, not reversible obstruction.
C	Correct	A large post-bronchodilator improvement shows reversibility.
D	Inverted operation	COPD shows persistent limitation.

12. Answer C Concept 8

A patient ~~who works as a pilot~~ has fatigue, low blood pressure, darkening skin, serum sodium 128 mEq/L, and potassium 5.9 mEq/L. ~~He recently ran a half marathon.~~ Which condition fits?

Target word: adrenal condition **Decides:** Low BP, low Na, high K, hyperpigmentation.

A	Inverted operation	Excess cortisol raises BP and glucose.
B	Camouflage	The race does not explain hyperkalemia and hyperpigmentation.
C	Correct	Cortisol and aldosterone loss cause this pattern.
D	Confusion set	Causes low sodium but not high potassium or hyperpigmentation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Left vs. right heart failure	Left-sided failure backs blood into the lungs (dyspnea, orthopnea, crackles); right-sided failure backs blood into the body (JVD, peripheral edema, hepatomegaly).	Linking leg edema to left-sided failure or crackles to right-sided failure.	Lung findings vs. systemic venous findings.
2	Acid-base interpretation	pH sets acidosis or alkalosis; if PaCO ₂ moves opposite to pH the cause is respiratory; if HCO ₃ moves with pH the cause is metabolic; the other value moving to offset it is compensation.	Naming the cause from whichever value is most abnormal.	An ABG with pH, PaCO ₂ , and HCO ₃ .
3	SIADH vs. diabetes insipidus	SIADH = too much ADH: water retention, low serum sodium, concentrated urine. DI = too little ADH effect: large volumes of dilute urine, high serum sodium.	Reversing which one causes hyponatremia.	Serum sodium and urine concentration together.
4	Anemia by red cell size	Iron deficiency causes microcytic anemia (low MCV); B12 or folate deficiency causes macrocytic anemia (high MCV), and B12 deficiency can add neurologic signs.	Assuming all anemia is iron deficiency.	MCV value plus neurologic or diet clues.
5	Shock types	Hypovolemic = low volume; cardiogenic = pump failure with high filling pressures; distributive (septic, anaphylactic, neurogenic) = vasodilation with low SVR; obstructive = blocked flow.	Treating every low blood pressure as volume loss.	Filling pressures, SVR, and skin warmth.
6	DKA vs. HHS	DKA: absolute insulin lack, ketones, metabolic acidosis, glucose often >250. HHS: some insulin present, very high glucose (often >600), high osmolality, little or no ketosis.	Expecting ketoacidosis in every hyperglycemic crisis.	pH, ketones, glucose level, and osmolality.
7	Asthma vs. COPD	Asthma causes variable, largely reversible airflow limitation; COPD causes persistent airflow limitation (post-bronchodilator FEV ₁ /FVC < 0.70).	Diagnosing COPD from wheezing alone.	Response to a bronchodilator on spirometry.

#	Concept	Rule	Trap	Tell
8	Cortisol excess vs. adrenal insufficiency	Cushing syndrome (excess cortisol): hyperglycemia, hypertension, central obesity, thin skin. Primary adrenal insufficiency (Addison): hypotension, hyponatremia, hyperkalemia, hyperpigmentation.	Mixing up which one raises potassium or blood pressure.	Blood pressure, sodium, potassium, and skin color.

Course scope (public description): Advanced Pathophysiology for the Advanced Practice Nurse prepares the graduate nursing student for the role of an advanced practice nurse with the competencies and skills needed to recognize disease states, identify disease progression, and assess and evaluate symptoms for patients across the lifespan. This course will help the graduate nursing student gain a deeper understanding of pathophysiology from the cellular to the systems level and will provide graduate nursing students with the knowledge and skills to determine the etiology, underlying physiological changes, and the human affective responses to alterations in health. This course will also prepare the graduate nursing student to communicate the pathophysiology of disease processes to providers and patients.